

Chapter 7

Money and Assets

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Nominal Rigidities and Market Power

- › The next section of the course deals with frictions that arise from nominal variables
- › As we move into GE, we'll see the impact of monopolistic pricing on inflation
- › We will study the fundamentals of New Keynesian analysis

Scenario

- › Imagine you are a vegetable farmer
- › You need a new pair of shoes
- › How do you get your shoes?

Double Coincidence of Wants

- › If the shoemaker is willing to accept vegetables, we have a **double coincidence of wants**
 - Two parties individually want what the other is offering
- › This example is one of a **barter economy**
 - Goods and services are traded for each other directly
 - NOT with money
 - Further, a **gift economy** is one where HHs gift each other resources with a social thought of reciprocity

So, What is Money?

- › **Money** is any resource that can be *readily* exchanged for goods and services
- › “Readily” is vague, but also meaningful
 - I can trade some oranges for pencils, but most people won't readily accept oranges as payment
 - Almost everyone in the U.S. will accept dollars

Extended scenario

- › The shoemaker is willing to accept wine for shoes
 - You have to go to a vineyard to see what they'll trade for wine
 - Assuming that's successful, the shoemaker has to go to their dye suppliers and find what they want as compensation...

- › Voluntary trade increases total surplus (Ricardian model)
 - Do we want to rely on double coincidence of wants to facilitate trade?

Yap (Waqab)

- › Small island group in the Pacific
 - First European contact in 1500's
 - Used limestone deposits on an island hundreds of miles away as money (rai)
 - Carved into stone discs & sailed back on small boats



Yap Proper: Rumung, Maap, Gagil-Tamil, and Marbaa'.

Credit

- › Given the inconvenience of moving stone, it became more abstract
 - Development of a credit system
 - People would keep score of who paid whom and tallies of limestone stock, making it easier to use for smaller transactions
 - "They often talk about the stones themselves not changing hands at all. In fact, most of the time they wouldn't." – Dr. Fitzpatrick
 - (Goldstein & Kestenbaum, 2010), (Fitzpatrick, 2004)

Value of Money

- › Society *must* buy into the item used as money
 - If enough people decide to stop valuing the stone, it becomes worthless

- › But the opposite is also true
 - If enough people value something, it can become money
 - Ex: Kent cigarettes in the final decade of communism in Romania
 - (Thurow, 1986)

3 Properties of Money

1. Medium of exchange
2. Unit of account
3. Store of value

Medium of Exchange

- › Money can be exchanged for goods and services
 - \$20 can buy many peanuts



"Boy-Scoutz 'n the Hood." *The Simpsons*, 1993.

Unit of Account

- › Provides a denomination to judge value
 - Numeraire property
 - Apple = \$1, melon = \$2
 - Melon is worth 2x as much as an apple
- › If apples were the **numeraire good**, the [relative] price of melons would be 2
 - Transactional reference point

Store of Value

- › Value holds in the future
 - \$10 today is \$10 tomorrow
 - The number 10 does not change
 - Not a perfect store of value if inflation exists...

Real and Nominal Variables

- › In Ch. 3 we laid out the basics of real and nominal GDP
- › Now, we'll see how the two concepts are connected
- › Ex: savings accounts do not quote your interest in units of apples today vs apples tomorrow, but in a percent of money saved
 - This is a nominal variable

The Link to Real

- › Variables: r = real interest, i = nominal interest, π = inflation
- › Let's “factor out” inflation from nominal interest to get real:

$$1+r = \frac{1+i}{1+\pi'}$$

- $(1 + \pi')$ is the factor of how much prices will change at end of loan

- › Then: $i = r + \pi' + r\pi'$

The Fisher Equation

- › With $i = r + \pi' + r\pi'$, in modern economies $r\pi'$ is very small
 - Around 0.001
 - Just drop it
- › Fisher Equation for nominal interest rate:

$$i = r + \pi'$$

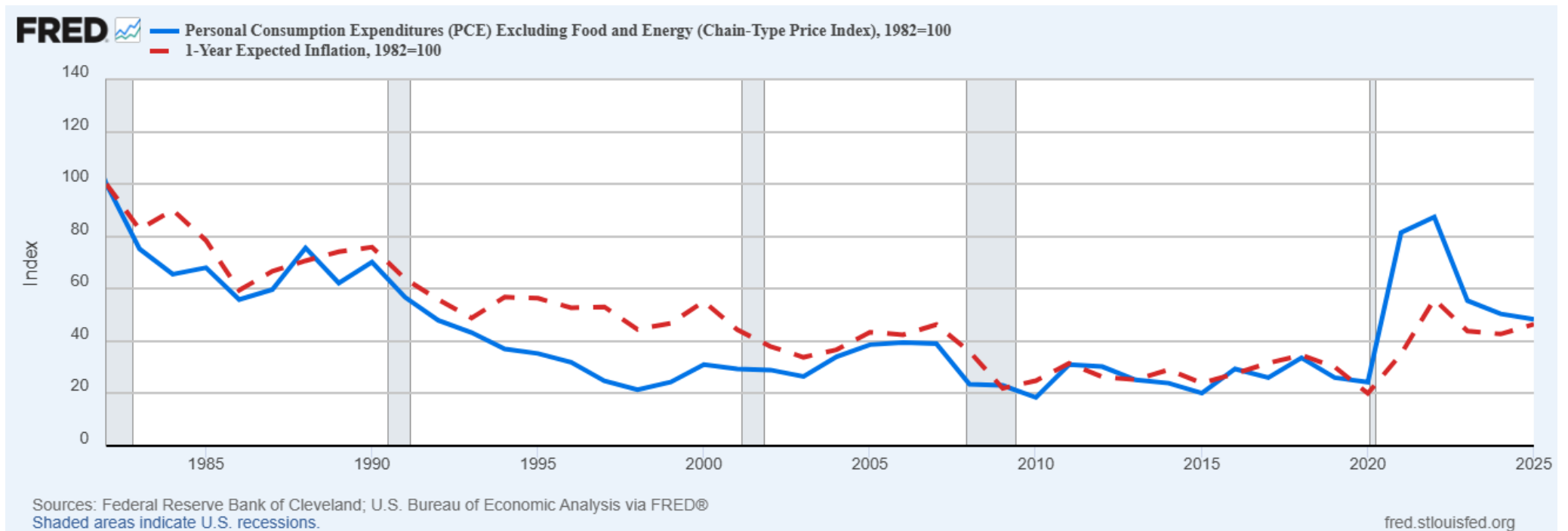
- › Which is great, but doesn't really get at HH behavior...

Expected Inflation

- › People write contracts for what they *expect* inflation to be
 - Expected inflation = $E[\pi]$
 - Expected inflation Fisher Equation: $i = r + E[\pi]$
- › Captures idea that people are “forward looking”
 - But are they accurate?

π

Expected Inflation Data



Does This Change Anything We've Done?

- › Nope! - HH's view π' of takes a little time to update
 - Initial change is just i
 - SE and WE still hold

- › Type of **adaptive expectations**
 - HH relies of pervious info to form future expectations

- › Firms still invest less when i increases
 - Cost of borrowing still goes up

But What about ***MB=MC***?

- › Convenience of single-output economy:
 - Multiply quantities by P to get their nominal value
 - $P \times Y$ = nominal value of output
 - $P \times MPN$ = nominal MPN (Marginal Revenue Product, $MRPN$)

$$P \times MPN = P \times w$$

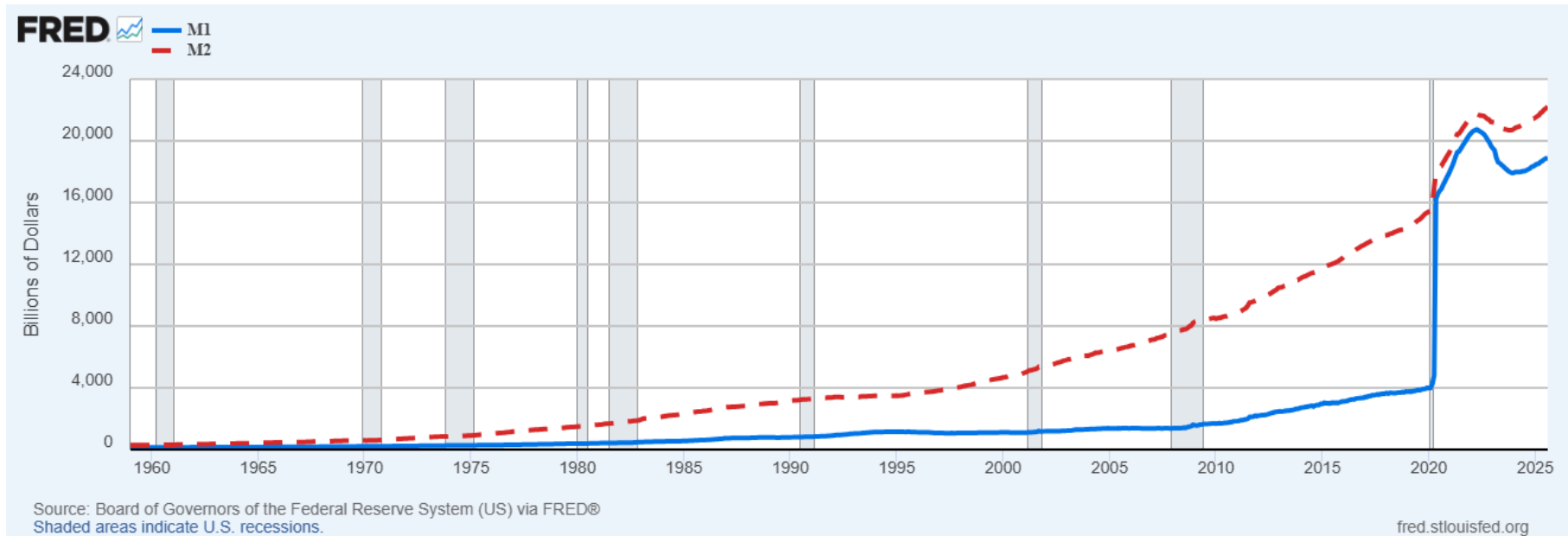
Two Types of Money

1. **Fiat money:** established by some central authority
 - Has no value on its own; basically just paper
 - Value based on trusting issuer and national economy
 - U.S. dollar is a fiat currency, printed by the Treasury

2. **Commodity money:** value comes from a commodity
 - Gold standard is the most well known
 - Incredibly useful commodity beyond jewelry
 - U.S. dollars represented a fixed amount of gold (and silver)
 - › 1933: U.S. removed gold standard domestically
 - › 1971: U.S. removed gold backing for international exchanges

The Money Supply

- › The **Federal Reserve (Fed)** controls the money supply
 - **M1**: Cash + liquid deposits (easily converted to cash)
 - **M2**: M1 + money market funds + time deposits



The Federal Reserve

- › The Fed is the monetary authority of the U.S.
- › The Fed influences the money supply and interest rates by:
 1. Open market operations (OMO) – today (Ch. 7)
 2. Setting reserve requirements – Ch. 8
 3. Quantitative easing – Ch. 8
 4. Simply picking an interest rate – Ch. 8

Open Market Operations

- › The Fed uses printed money to:
 - Increase M^s by purchasing assets from the public
 - Decrease M^s by selling assets to the public

- › Simply moves M^s left or right depending on what they want

Nominal Money Demand

- › Demand for money M^d in nominal terms is a function of:
 - Price level (P)
 - Nominal interest rate (i)
 - Lifetime resources (Ω)

$$M^d = P \times L(\Omega, i)$$

- where $L()$ is a preference function for Liquid money

Lifetime Resources

› Lifetime resources (Ω) are divided into:

1. Value of non-liquid assets: qB

› q is the price of a composite (basket) asset, B

▪ HH can choose B

2. Value of lifetime income: $h(Y, r)$

▪ HH cannot choose Y (or at least has negligible influence over)

› Similar to borrowing against future income in last chapter

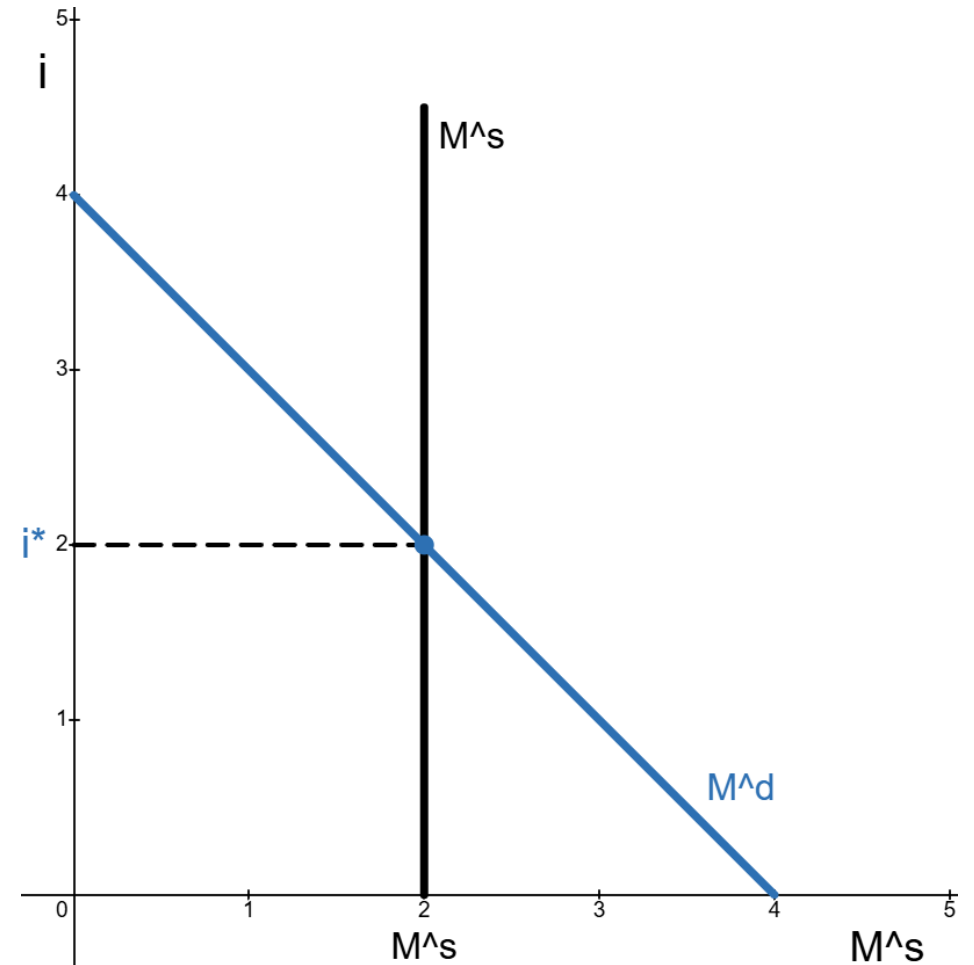
› **$\Omega = qB + h(Y, r)$**

The Effect of Ω on M^d

- › When the HH's resources, Ω , grow so does M^d
- › If $Y \uparrow$, then $\Omega \uparrow$:
 - $M^d \uparrow$ since HHs use cash for c (medium of exchange)
 - M^d is therefore a **derived demand**
 - › Comes from demand of another good (c)
 - Wealth effect encouraging $c \rightarrow$ HH needs more cash

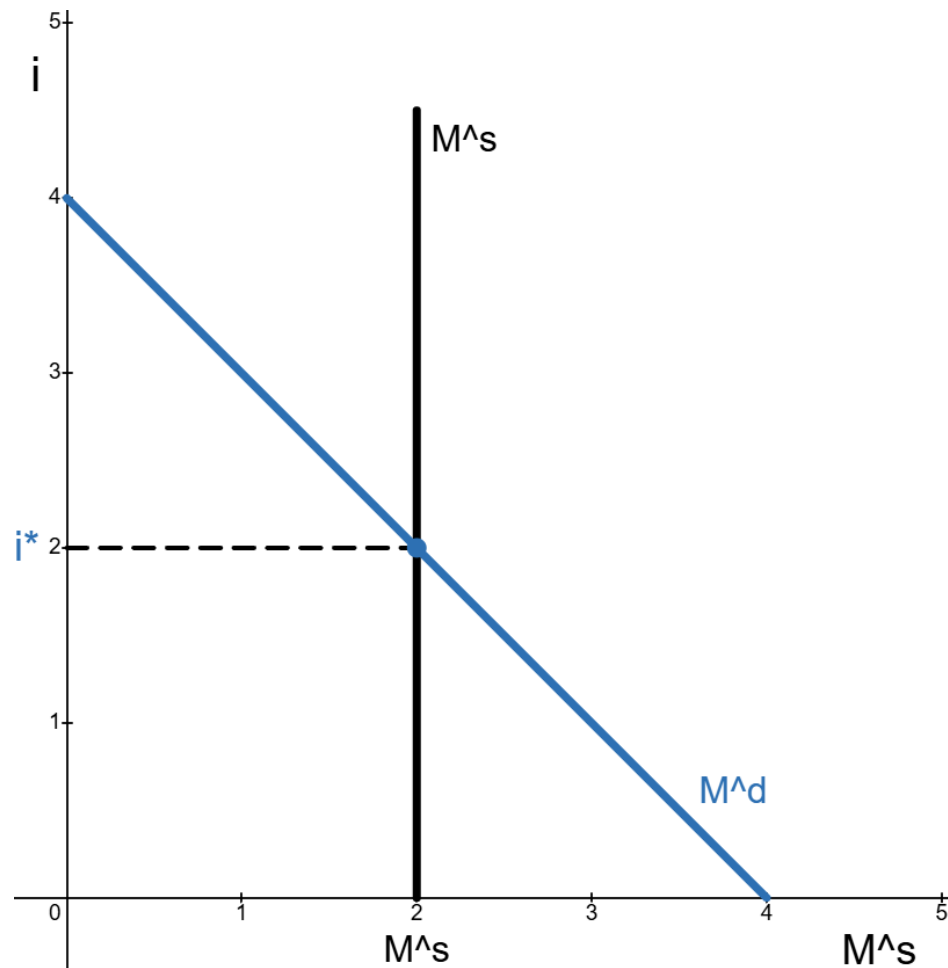
Graphing the Money Market

- › M^s is simply set by the Fed
- › M^d is decreasing in i
 - Why?



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Increase in M^s



Inflation and Real Money Demand

- › Inflation adds onto the substitution effect
 - Increases opportunity cost of holding money
 - Value choice: \$3 gallon of milk today or \$4 gallon of milk tomorrow
 - Real money demand falls with higher expected inflation

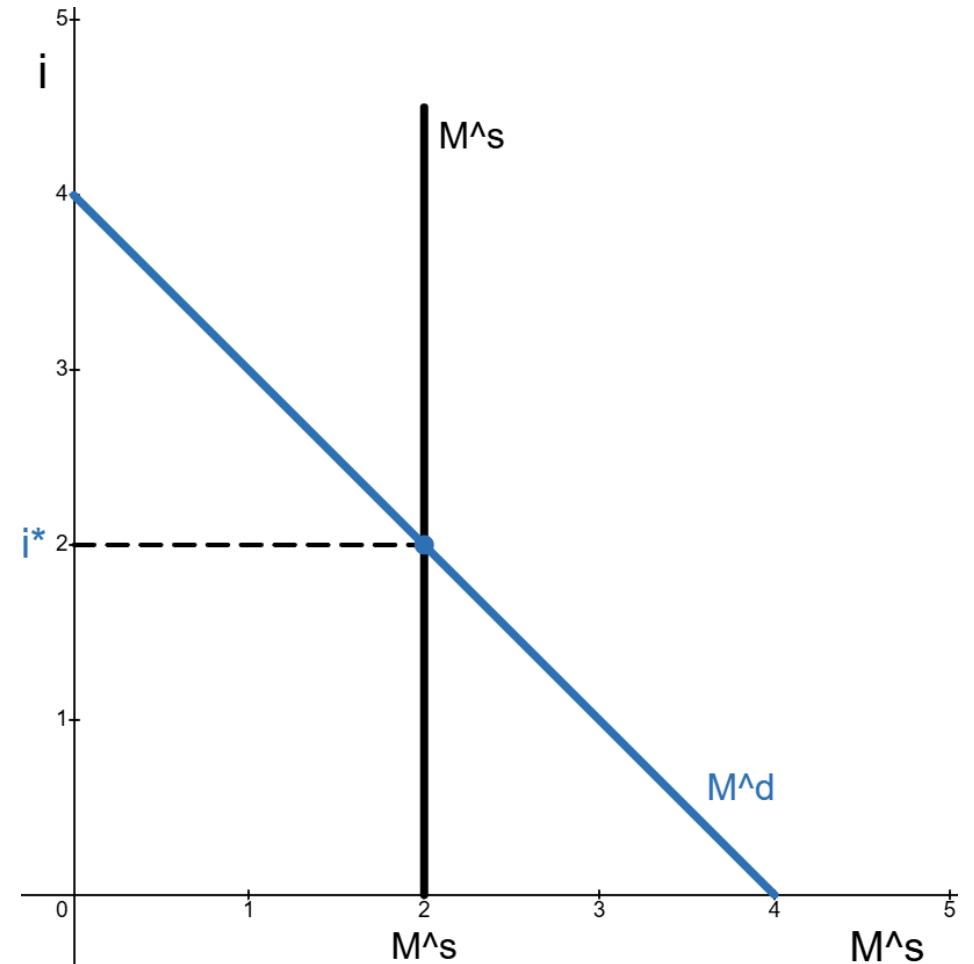
Price Level vs Inflation

- › Now that we have a function of P and π , we need to be a bit more detailed than in Ch. 3:

P	π	Definition
10.00	-	-
11.00	10%	Inflation
12.10	10%	Inflation (constant)
14.52	20%	Inflation (accelerating)
15.25	5%	Disinflation
14.94	-2%	Deflation

Changes in Y

- › Increase in Y raises demand for real money balances
- › Decreases in Y lower demand for real money balances
- › Both lead to a new equilibrium



Equilibrium in the Asset Market

- › Equilibrium in money market implies: $M^d = M^s$
 - And gives us i^*

- › HH “chooses” how much to save in B
 - But really, it’s just a residual
 - We don’t need to “solve” for B ; just falls out of HH’s budget

- › Walras Law: if $N-1$ markets clear, then N markets clear
 - (Walras, 1874)

A Note on i and q

- › Let B involve claims on the firm
- › Assume M^s increases so that i falls
 - Firms invest more; a share of the firm becomes *more* valuable
 - q rises when i falls
- › Therefore, i and q move in opposite directions
 - Further reading on related, but not identical concept, “Tobin’s q ”
 - In aggregate: ratio of market value to replacement cost
 - (Tobin & Brainard, 1976)

But Doesn't That Mean M^d Changes?

- › Yes! There are some feedback effects happening here
- › Just not enough to change our qualitative answer

References (1/2)

1. Board of Governors of the Federal Reserve System (US), M1 [M1SL], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/M1SL>, .
2. Board of Governors of the Federal Reserve System (US), M2 [M2SL], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/M2SL>, .
3. "Boy-Scoutz 'n the Hood." *The Simpsons*, season 5, episode 8 Fox, 18 Nov. 1993.
4. Federal Reserve Bank of Cleveland, 1-Year Expected Inflation [EXPINF1YR], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/EXPINF1YR>, .
5. Federal Reserve Bank of St. Louis, Velocity of M1 Money Stock [M1V], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/M1V>, .
6. Federal Reserve Bank of St. Louis, Velocity of M2 Money Stock [M2V], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/M2V>, .
7. Fisher, Irving. *The Rate of Interest: Its nature, determination and relation to economic phenomena*. Macmillan, 1907.
8. Fitzpatrick, Scott M. "Stone Money." 2004
9. Goldstein, Jacob, and David Kestenbaum. "The Island of Stone Money." *NPR*, NPR, 10 Dec. 2010, www.npr.org/sections/money/2011/02/15/131934618/the-island-of-stone-money.
10. Thurow, Roger. "In Romania, Smoking a Kent Cigarette is Like Burning Money." *Wall Street Journal* (1986).

References (2/2)

11. Tobin, James, and William C. Brainard. "Asset markets and the cost of capital." Cowles Foundation Discussion Paper No. 427, Cowles Foundation for Research in Economics, Yale University (1976).
 12. U.S. Bureau of Economic Analysis, Personal Consumption Expenditures Excluding Food and Energy (Chain-Type Price Index) [PCEPILFE], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/PCEPILFE>, .
 13. Walras, Léon. "Elements of Pure Economics." (1874).
- A. All non-cited graphs for this course made by me on [Desmos](#).

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APPENDIX

Real Money Demand and Equilibrium

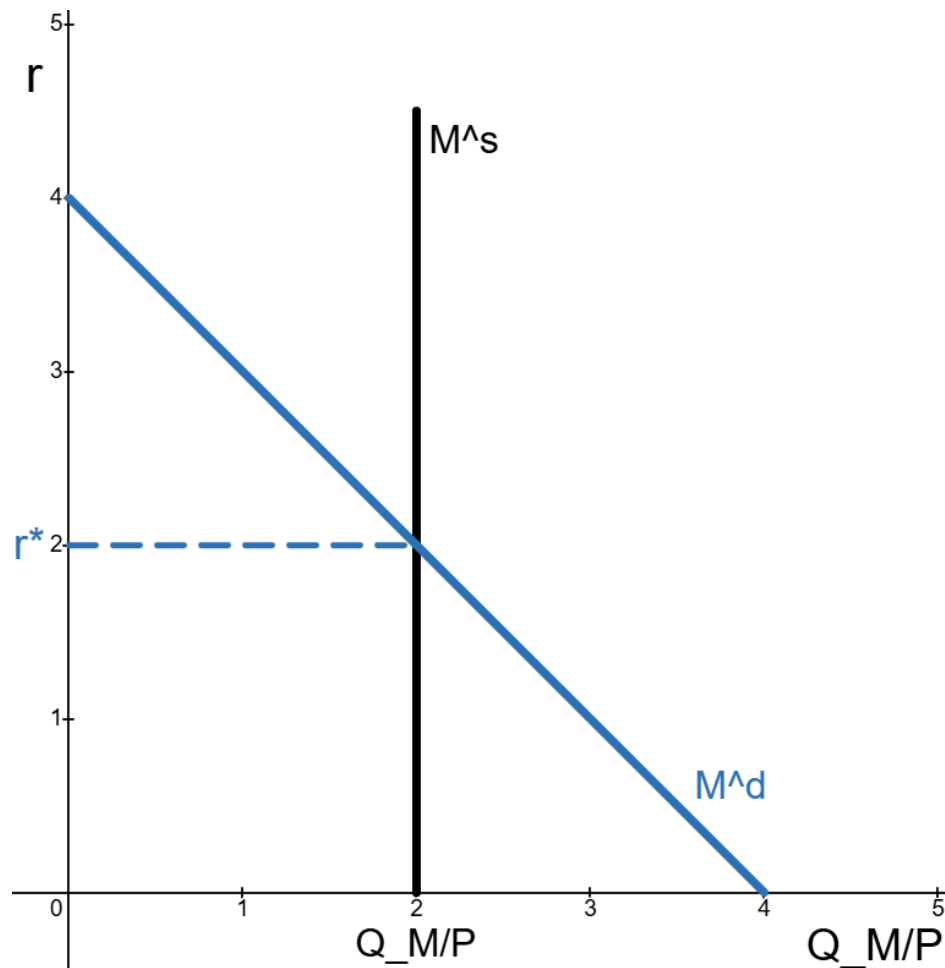
› We can divide by P (and redefine some things) to get:

$$\frac{M^d}{P} = L(\Omega, r + E[\pi])$$

› With the equilibrium condition being:

$$\frac{M^s}{P} = \frac{M^d}{P} = L(\Omega, r + E[\pi])$$

The Market for REAL Money Balances



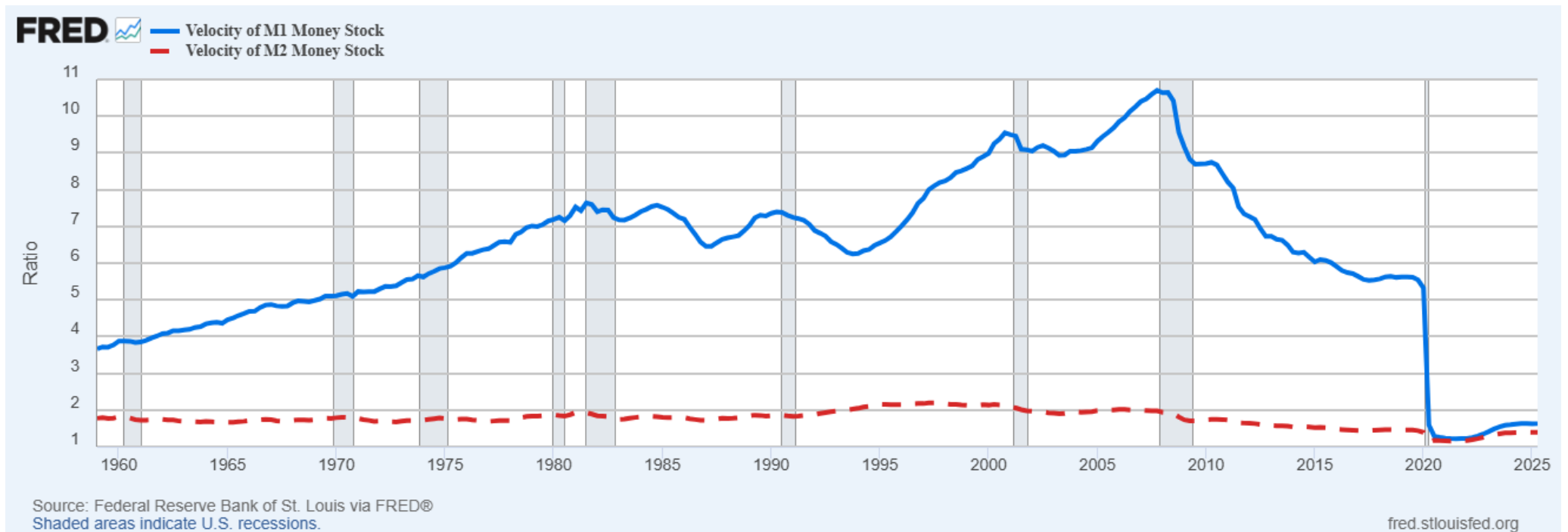
Velocity of Money

- › Related concept: **velocity (V)** of money
 - How often money changes hands

$$V = \frac{PY}{M^s}$$

- As V increases, each dollar is being used more
- › Assuming velocity is a constant, k : $\frac{M^d}{k} = PY$ and $V = \frac{1}{k}$
 - Quantity Theory of Money Cambridge equation

Velocity of M1 vs M2



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